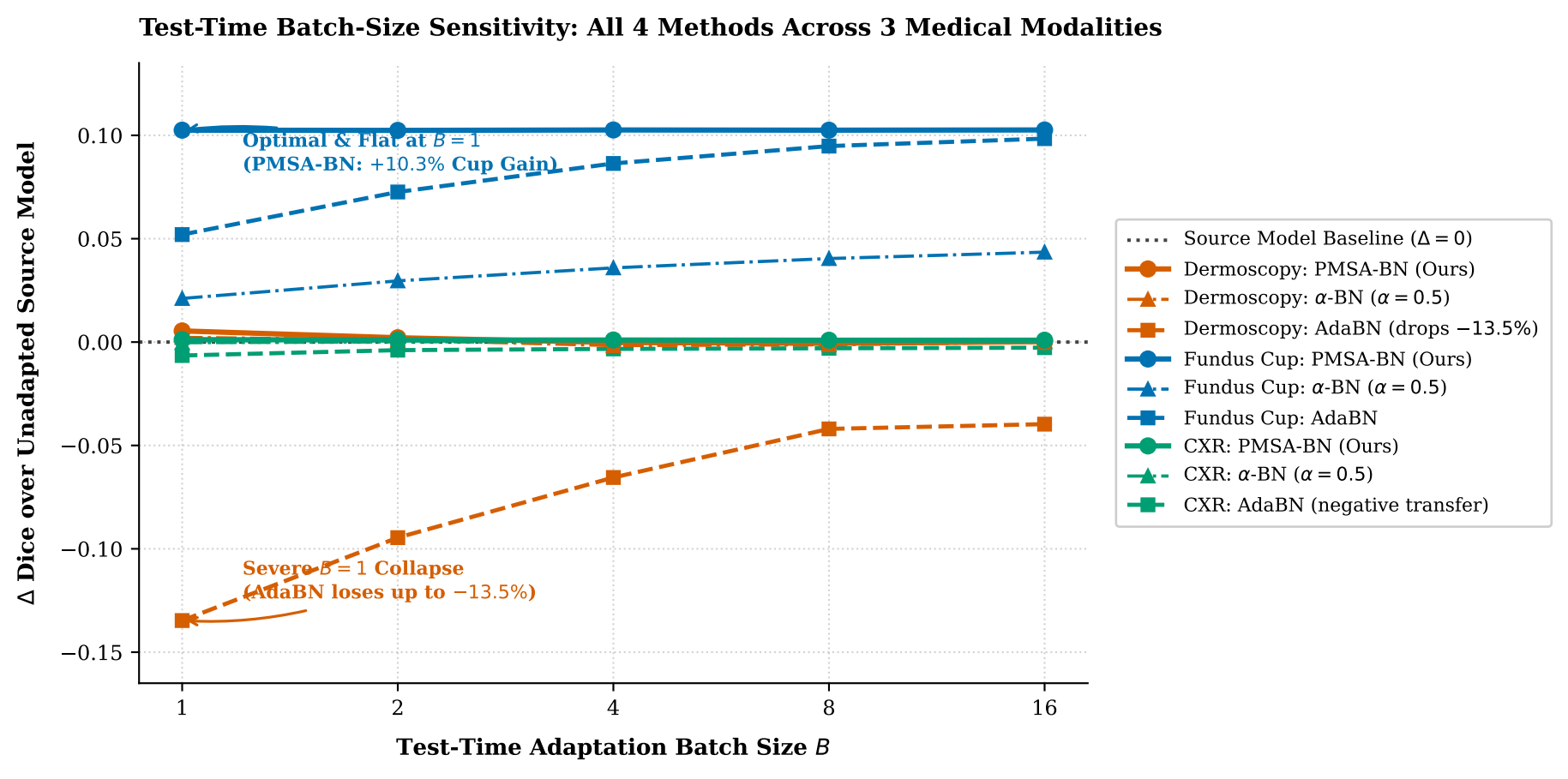
Test-Time Batch-Size Sensitivity: All 4 Methods Across 3 Medical Modalities
Δ Dice over Unadapted Source Model
0.10
0.05
0.00
−0.05
−0.10
−0.15
Optimal & Flat at B = 1
(PMSA-BN: +10.3% Cup Gain)
Severe B = 1 Collapse
(AdaBN loses up to −13.5%)
1
2
4
8
16
Test-Time Adaptation Batch Size B
Source Model Baseline (Δ = 0)
Dermoscopy: PMSA-BN (Ours)
Dermoscopy: α-BN (α = 0.5)
Dermoscopy: AdaBN (drops −13.5%)
Fundus Cup: PMSA-BN (Ours)
Fundus Cup: α-BN (α = 0.5)
Fundus Cup: AdaBN
CXR: PMSA-BN (Ours)
CXR: α-BN (α = 0.5)
CXR: AdaBN (negative transfer)